



Sydney Girls High School

2002

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

Extension 2

This is a trial paper ONLY. It does not necessarily reflect the format or the contents of the 2002 HSC Examination Paper in this subject.

General Instructions

- Reading Time - 5 mins
- Working time - 3 hours
- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are supplied
- Board-approved calculators may be used.
- Diagrams are not to scale
- Each question attempted should be started on a new sheet. Write on one side of the paper only.

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Candidate Number

Question One (15 marks)

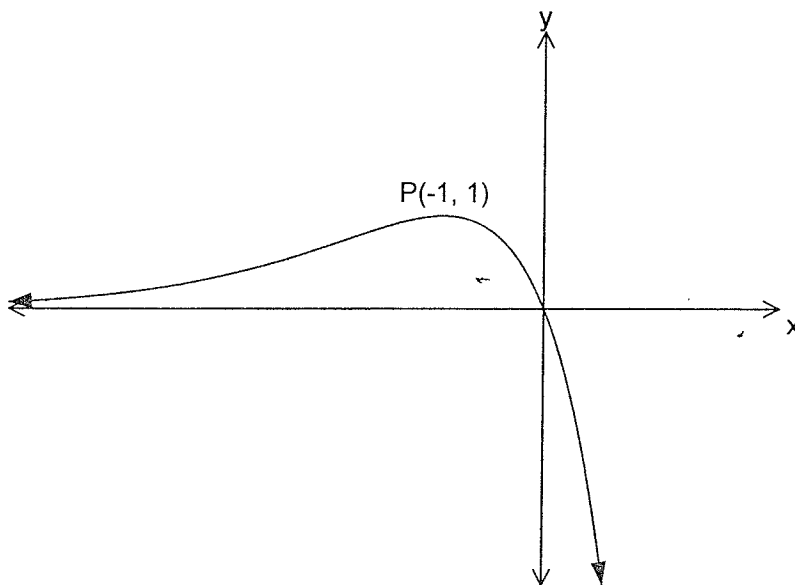
1. Find $\int \frac{1}{x} (1 + \log_e x)^4 dx$ [2]
2. Express $\frac{1}{x^2 + 3x - 4}$ in the form $\frac{A}{x+4} + \frac{B}{x-1}$, hence find $\int \frac{dx}{x^2 + 3x - 4}$ [3]
3. Find $\int \sin^3 \theta \cos^2 \theta d\theta$ [3]
4. Find $\int \frac{2x+5}{x^2 + 4x + 5} dx$ [3]
5. Find $\int e^{2x} \sin 3x dx$ [4]

Question Two (15 marks)

1. Given $z = 2 - 3i$ [3]
 - a) Find $\frac{1}{z}$
 - b) Find iz
 - c) Give a geometrical interpretation of your answer to part b)
2. Find real number x and y such that $3x + 2iy - ix + 5y = 7 + 5i$ [2]
3. Given $z = 4 + 4\sqrt{3}i$ [3]
 - a) Find $|z|$ and $\arg z$
 - b) Hence evaluate $(4 + 4\sqrt{3}i)^9$
4. Sketch the locus given by $\arg\left(\frac{z+i}{z-i}\right) = \frac{\pi}{2}$ [2]
5. Find the Cartesian equation of the curve represented by $\frac{(z + \bar{z})}{2} = |z| - 2$ [2]
and describe it.
6.
 - a) Solve the equation $z^3 - 1 = 0$ giving your answers in modulus-argument form. [3]
 - b) Let w be the root of $z^3 - 1 = 0$ with the smallest positive argument
 - i) Show $1 + w + w^2 = 0$
 - ii) Simplify $(1 + w^2)^4$

Question Three (15 marks)

1. Given $y = f(x) = (x-2)^2(x+1)$ sketch without using calculus [5]
- a) $y = f(x)$
- b) $y = \frac{1}{f(x)}$
- c) $y^2 = f(x)$
2. The graph of $y = F(x)$ is shown below. The graph has a maximum turning point at $P(-1, 1)$. [7]



Sketch on separate diagrams showing all relevant features

- a) $y = F(-x)$
- b) $y = \log_e [F(x)]$
- c) $y = e^{F(x)}$
- d) $y = [F(x)]^2$
3. Sketch the graph of $x^3 + y^3 - 1 = 0$ [3]

Question Four (15 marks)

1. The equation of a conic is $\frac{x^2}{4} + \frac{y^2}{3} = 1$ find [4]

- a) The eccentricity
- b) The co-ordinates of the foci
- c) The equations of the directrices
- d) Sketch the conic showing vertices foci and directrices.

2. Find the equation of the chord of contact of the tangents to the hyperbola $x^2 - 16y^2 = 16$ from the point with coordinates (2, -4) [2]

3. Find the equation of the hyperbola with foci at $(\pm 5, 0)$ and eccentricity $e = \frac{5}{4}$ [3]

4. The point $P(5\cos\theta, 3\sin\theta)$ lies on the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. [6]

The normal at P crosses the x-axis at Q and the y-axis at R .

- a) Derive the equation of the normal at P
- b) Show that the midpoint M of QR lies on the ellipse with

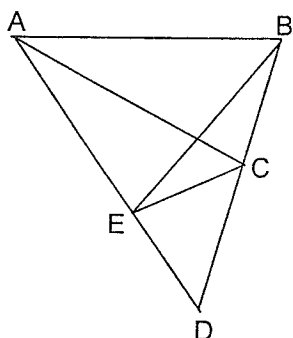
equation $\frac{25x^2}{64} + \frac{9y^2}{64} = 1$

Question Five (15 marks)

1. a) Sketch the graph of $g(x) = 1 + \frac{1}{x+1}$ for $x > -1$ [4]
 b) Find the equation of the inverse $g^{-1}(x)$ and sketch it on a separate set of axes.
 c) Solve $g(x) = g^{-1}(x)$

2. If α, β and γ are the roots of the equation $x^3 - 2x^2 + 4x + 2 = 0$, [2]
 Find the equation, which has roots
 a) $(\alpha-1), (\beta-1)$ and $(\gamma-1)$
 b) $\frac{\alpha}{2}, \frac{\beta}{2}$ and $\frac{\gamma}{2}$

3. In the figure below $\angle AEB = \angle BCA$ [2]



Prove $\angle BAE = \angle ECD$

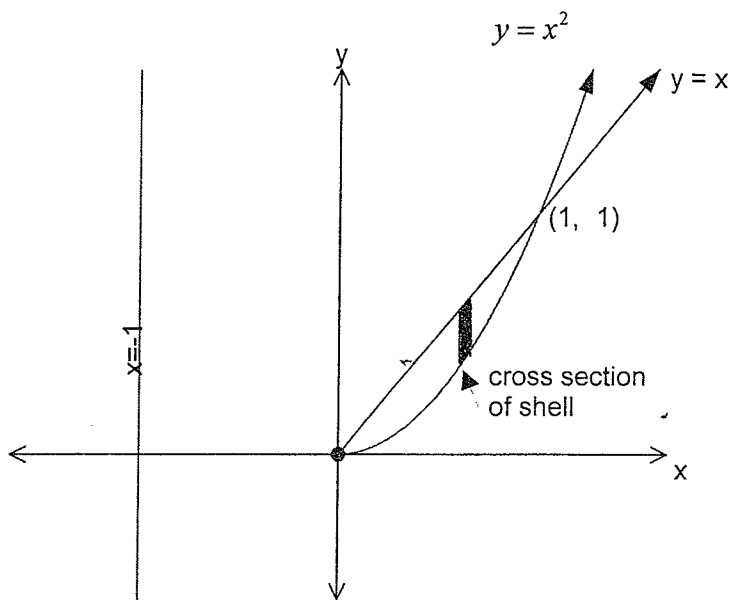
4. Given that the equation $x^4 - 2x^3 - 12x^2 + 40x - 32 = 0$ has a triple root, [3]
 find all the roots of this equation.
5. A solid has its base area bounded by $y = \sin x$, the x-axis, [4]
 $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$ Each cross section perpendicular to the x-axis is a square
 with one side on the base. Find the volume of the solid.

Question Six (15 marks)

1. $P(x)$ is an even monic polynomial of degree four with integer coefficients. One zero is $3i$ and the product of the zeros is -18 . Factorise $P(x)$ fully over the real field. [3]

2. Prove that $\frac{\cos 15^\circ + \cos 75^\circ}{\sin 15^\circ - \sin 75^\circ} = -\sqrt{3}$ [3]

3. [6]



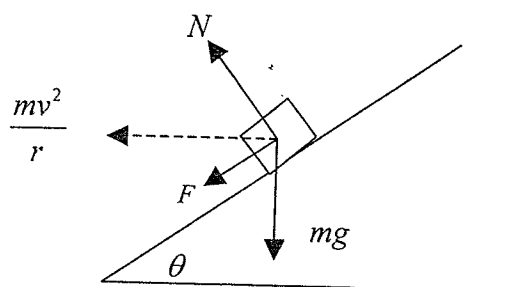
Use the method of cylindrical shells to calculate the volume of the solid formed when the area bounded by $y = x$ and $y = x^2$ is rotated about the line $x = -1$

4. Find and sketch the locus of z if $|w| = 1$ and $z = \frac{w+7}{1-w}$ [3]

Question Seven (15 marks)

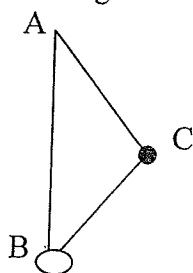
1. A train line banked at an angle θ as shown below.

[8]



- a) If the force of circular motion is given as $\frac{mv^2}{r}$, and the force due to gravity as mg , determine the components of frictional force F and the normal reaction N in terms of m , g , v , r and θ
- b) A train turning a corner of radius 500 metres causes the same frictional force along the slope travelling at 30 km/h as it does travelling at 90 km/h. (Note the two frictional forces are in different directions but are the same magnitude)
- Find the angle at which the track is banked (answer to the nearest minute)
 - Find the speed in km/h for which the frictional force is negligible (answer to the nearest km/h)
2. A four metre string attached at A has a 1kg mass attached at C and a 2kg metal ring attached at B . The ring at B is free to slide up and down the smooth vertical light rod descending from A .

[7]

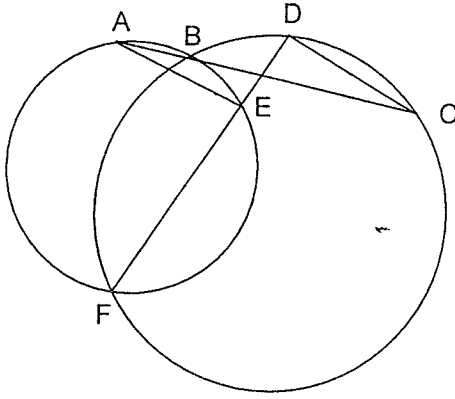


Given $AC = BC = 2$ metres and $\angle BAC = 30^\circ$

- Find the angular velocity of the mass at C , which is rotating about the rod AB so that the ring at B remains stationary.
- If the mass at C is changed to a 3kg mass and retains the same angular velocity as in part a) find;
 - The new size of $\angle BAC$ (answer to the nearest degree)
 - How far up the smooth rod the ring at B will rise before becoming stationary. (Answer to the nearest cm)

Question Eight (15 marks)

1. Use mathematical induction to prove that $x^{2n} - y^{2n}$ is divisible by $(x + y)$ for $n \geq 1$ (n an integer) [4]
2. If p and q are the roots of $\frac{1}{x} + \frac{1}{x+a} + \frac{1}{x+b} = 0$ and given that $a^2 + b^2 = 4ab$ prove that $p^2 + q^2 = 6pq$ [4]
3. In the figure below ABC and DEF are straight lines. Prove AE parallel to DC [3]



4. Prove $|z-1| + |z+1| \leq 2\sqrt{2}$ given that $|z| \leq 1$ [4]

$\alpha\beta\chi\delta\epsilon\phi\varphi\eta\iota\kappa\lambda\mu\nu\pi\theta\vartheta\rho\sigma\varsigma\tau\upsilon\omega\xi\psi$

Question One

$$1. I = \int \frac{1}{5x} (1 + \log_e x)^4 dx$$

$$\text{let } u = 1 + \log_e x$$

$$du = \frac{dx}{x}$$

$$I = \int u^4 du$$

$$= \frac{1}{5} u^5 + C$$

$$= \frac{1}{5} (1 + \log_e x)^5 + C \quad (2)$$

$$2. \frac{1}{x^2 + 3x - 4} = \frac{A}{x+4} + \frac{B}{x-1}$$

$$1 \equiv A(x-1) + B(x+4)$$

$$\text{put } x=1, \quad B = \frac{1}{5}$$

$$\text{put } x=-4, \quad A = -\frac{1}{5}$$

$$\int \frac{dx}{x^2 + 3x - 4} = \frac{1}{5} \int \frac{dx}{x-1} - \frac{1}{5} \int \frac{dx}{x+4}$$

$$= \frac{1}{5} \log_e (x-1) - \frac{1}{5} \log_e (x+4) + C$$

$$= \frac{1}{5} \log_e \left(\frac{x-1}{x+4} \right) + C \quad (3)$$

$$3. I = \int \sin^3 \theta \cos^2 \theta d\theta$$

$$= \int (1 - \cos^2 \theta) \cos^2 \theta \sin \theta d\theta$$

$$\text{put } u = \cos \theta, \quad du = -\sin \theta d\theta$$

$$I = -\int (1 - u^2) u^2 du$$

$$= -\int u^2 - u^4 du$$

$$= -\frac{1}{3} u^3 + \frac{1}{5} u^5 + C$$

$$= \frac{1}{5} \cos^5 \theta - \frac{1}{3} \cos^3 \theta + C \quad (3)$$

$$4. I = \int \frac{2x+5}{x^2+4x+5} dx$$

$$= \int \frac{2x+4}{x^2+4x+5} dx + \int \frac{1}{x^2+4x+4+1} dx$$

$$= \int \frac{2x+4}{x^2+4x+5} dx + \int \frac{dx}{(x+2)^2+1}$$

$$= \log_e (x^2+4x+5) + \tan^{-1} (x+2) + C \quad (3)$$

$$5. I = \int e^{2x} \sin 3x dx$$

$$\text{let } u = \sin 3x, \quad v = e^{2x}, \quad u' = 3 \cos 3x, \quad v' = \frac{1}{2} e^{2x}$$

$$I = \frac{1}{2} e^{2x} \sin 3x - \frac{3}{2} \int e^{2x} \cos 3x dx$$

$$= \frac{1}{2} e^{2x} \sin 3x - \frac{3}{2} I_1$$

$$\text{let } u = \cos 3x, \quad v = e^{2x}, \quad u' = -3 \sin 3x, \quad v' = \frac{1}{2} e^{2x}$$

$$I_1 = \frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} \int e^{2x} \sin 3x dx$$

$$= \frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} I$$

$$\therefore I = \frac{1}{2} e^{2x} \sin 3x - \frac{3}{2} \left[\frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} I \right]$$

$$= \frac{1}{2} e^{2x} \sin 3x - \frac{3}{4} e^{2x} \cos 3x - \frac{9}{4} I$$

$$\frac{13}{4} I = \frac{1}{2} e^{2x} \sin 3x - \frac{3}{4} e^{2x} \cos 3x$$

$$I = \frac{2}{13} e^{2x} \sin 3x - \frac{3}{13} e^{2x} \cos 3x$$

$$I = \frac{1}{13} (2e^{2x} \sin 3x - 3e^{2x} \cos 3x) + C \quad (4)$$

Question Two

$$1. z = 2 - 3i$$

$$a) \frac{1}{2-3i} \times \frac{2+3i}{2+3i} = \frac{1}{13} (2+3i)$$

$$b) iz = i(2-3i)$$

$$= 3 + 2i \quad (3)$$

c) rotation of z through 90° anticlockwise

$$2. 3x + 2iy - ix + 5y = 7 + 5i$$

$$3x + 5y + i(2y - x) = 7 + 5i$$

equate real, imaginary

$$3x + 5y = 7 \quad (1)$$

$$-x + 2y = 5 \quad (2)$$

$$6 \times 3 \quad -3x + 6y = 15 \quad (2)$$

$$(1) + (2) \quad 11y = 22$$

$$y = 2, \quad x = -1 \quad (2)$$

$$3. z = 4 + 4\sqrt{3}i$$

$$a) |z| = \sqrt{16 + 48}$$

$$= 8$$

$$b) \arg z = \tan^{-1} \frac{4\sqrt{3}}{4}$$

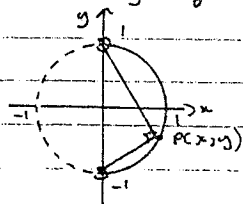
$$= \tan^{-1} \sqrt{3}$$

$$= \frac{\pi}{3}$$

$$\begin{aligned}
 \text{b)} \quad (4 + 4\sqrt{2}i)^9 &= \left(8 \operatorname{cis} \frac{\pi}{3}\right)^9 \\
 &= 2^{27} \operatorname{cis} 3\pi \\
 &= 2^{27} (-1) \\
 &= -2^{27}
 \end{aligned}$$

$$4. \arg\left(\frac{3+i}{3-i}\right) = \frac{\pi}{2}$$

$$\text{ie } \arg(3+i) - \arg(3-i) = \frac{\pi}{2}$$



$$5. \frac{1}{2} [x+iy + x-iy] = \sqrt{x^2+y^2} - 2$$

$$x+2 = \sqrt{x^2+y^2}$$

$$x^2 + 4x + 4 = x^2 + y^2$$

$$y^2 = 4(x+1) \quad \text{ie parabola vertex } (-1, 0)$$

Major Axis $\times$ focus $(0, 0)$

$$6. \text{a)} \quad z^3 = 1$$

$$= \operatorname{cis} 0$$

$$z^k = \operatorname{cis} (2\pi k/3), \quad k = 0, 1, 2$$

$$z = \operatorname{cis} \left(\frac{2\pi k}{3}\right)$$

$$\begin{aligned}
 z_0 &= \operatorname{cis} 0 = 1, \quad z_1 = \operatorname{cis} \frac{2\pi}{3} = w, \quad z_2 = \operatorname{cis} \frac{4\pi}{3} = w^2 \\
 &= 1, \quad = w, \quad = w^2
 \end{aligned}$$

$$\text{b) i) } 1 + w + w^2 = \text{sum of roots}$$

$$= -\frac{0}{1} \quad \left(-\frac{b}{a}\right)$$

$$= 0 \quad [\text{or sum c.p.}]$$

$$\text{ii) } (1+w)^4 = (-w)^4$$

$$= w^4$$

$$= w^3 \cdot w$$

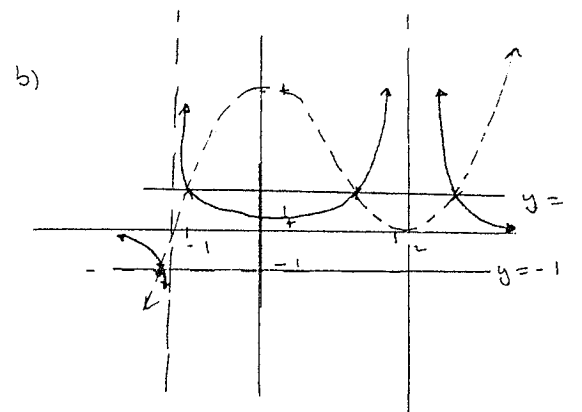
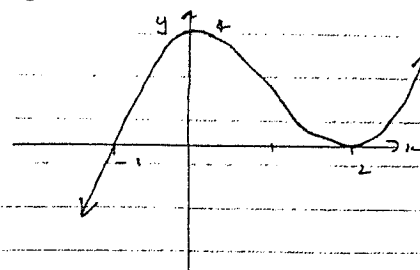
$$= w$$

$$\text{since } w^3 = 1$$

$$\therefore w^3 = 1$$

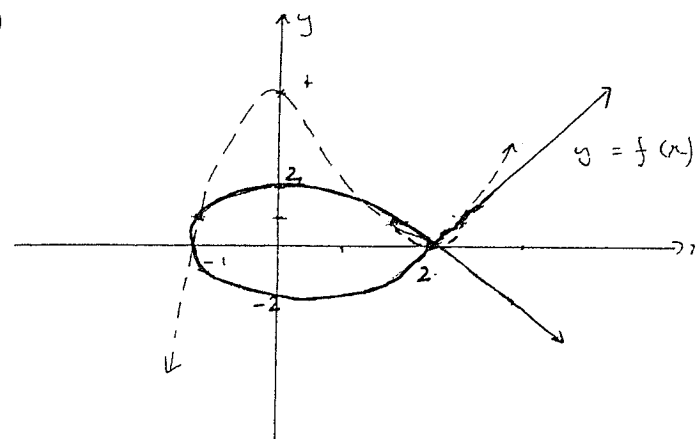
Question Three

$$1. \text{a) } y = f(x) = (x-2)^2(x+1)$$



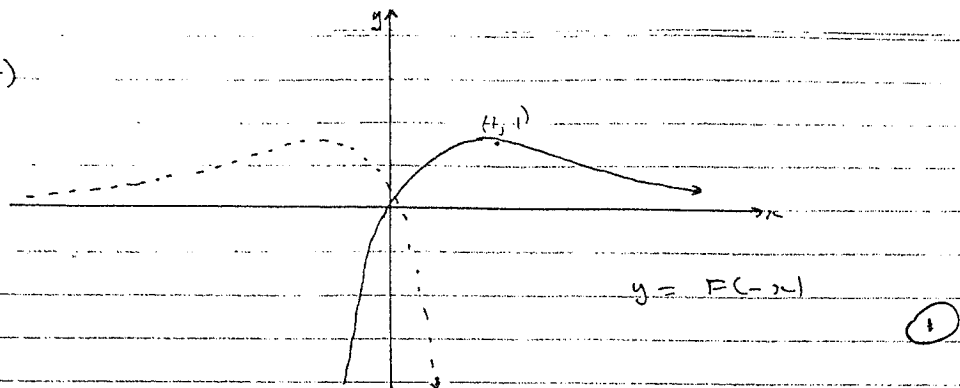
$$y = \frac{1}{f(x)}$$

c)

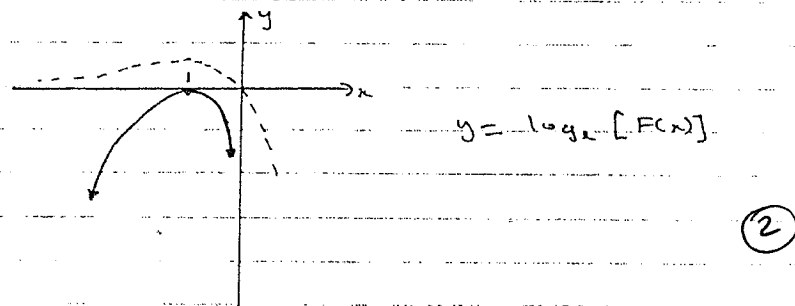


$$y = \sqrt{f(x)}$$

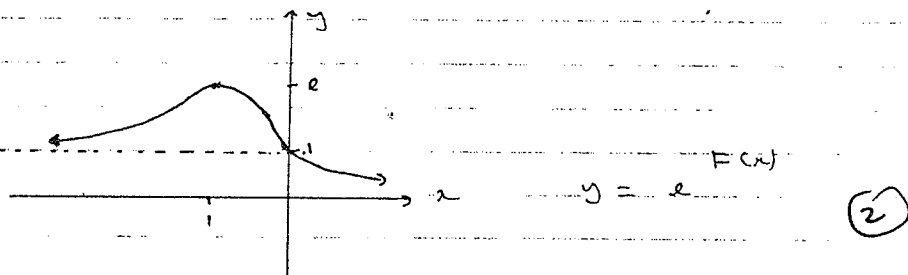
2. a)



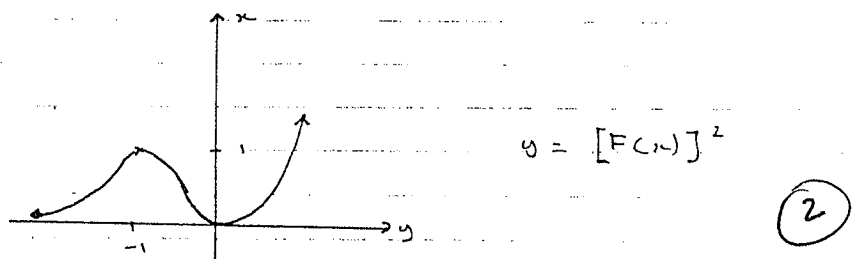
b)



c)



d)

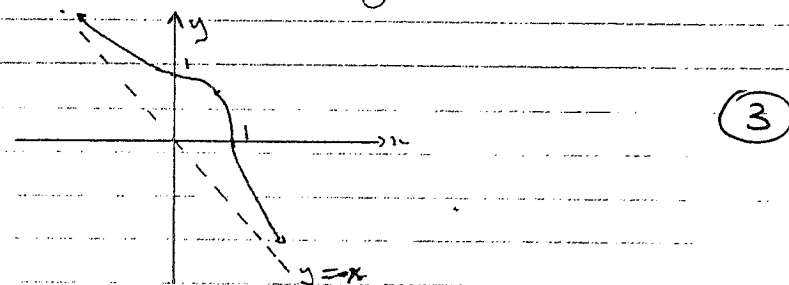


3. $x^2 + y^2 - 1 = 0$ Intercepts $x=1$, $y=1$

as $x \rightarrow \infty$ $y \rightarrow -x$. asymptote: $y = -x$
 x and y can be interchanged $\therefore$ symmetrical $y = x$

Also, consider: $3x^2 + 3y^2 \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} = -\frac{x^2}{y^2}$ when $x=0 \rightarrow$ horizontal tan
 when $y=0 \rightarrow$ vertical tan



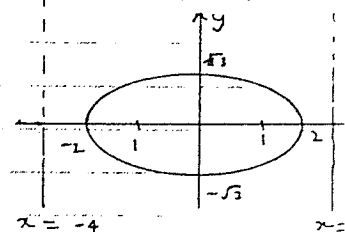
Question Four

1. $\frac{x^2}{4} + \frac{y^2}{3} = 1$ $a=2$, $b=\sqrt{3}$
 a) $b^2 = a^2(1-e^2)$ b) Foci $(\pm a, 0)$
 $3 = 4(1-e^2)$ $(\pm 1, 0)$

$$3 = 4 - 4e^2$$

$$-1 = -4e^2 \quad e = \frac{1}{2}$$

c) Directrices $x = \pm \frac{a}{e}$
 $= \pm 4$



2. $x^2 - 16y^2 = 16$
 $\frac{x^2}{16} - y^2 = 1$ $(2, -4)$ $a^2=16$, $b^2=1$

$$\frac{x x_0}{a^2} - \frac{y y_0}{b^2} = 1$$

$$\frac{2x}{16} + \frac{4y}{1} = 1$$

$$2x + 64y = 16$$

$$x + 32y = 8$$

$$3. \text{ Focus at } (-5, 0), e = \frac{5}{4}$$

$$ae = 5$$

$$a \cdot \frac{5}{4} = 5 \therefore a = 4$$

$$\text{and } b^2 = a^2 (e^2 - 1) \\ = 16 \left(\frac{25}{16} - 1 \right)$$

$$b^2 = 9$$

$$\text{ie } \frac{x^2}{16} - \frac{y^2}{9} = 1$$

(3)

$$4. a) \frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$\frac{2x}{25} + \frac{2y}{9} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-9x}{25y}$$

$$\text{at } P \quad m_1 = \frac{-45 \cos \theta}{75 \sin \theta} \\ = \frac{-3 \cos \theta}{5 \sin \theta}$$

$$\therefore m \text{ of normal } m_2 = \frac{5 \sin \theta}{3 \cos \theta}$$

$$\text{Eqn. of Normal } y - 3 \sin \theta = \frac{5 \sin \theta}{3 \cos \theta} (x - 5 \cos \theta)$$

$$3y \cos \theta - 9 \sin \theta \cos \theta = 5x \sin \theta - 25 \sin^2 \theta$$

$$\text{ie } 5x \sin \theta - 3y \cos \theta - 16 \sin \theta \cos \theta = 0$$

$$b) \text{ at } \theta \quad y = 0 \quad 5x \sin \theta - 16 \sin \theta \cos \theta = 0$$

$$x = \frac{16 \cos \theta}{5}$$

$$\text{ie } Q \left(\frac{16 \cos \theta}{5}, 0 \right)$$

$$\text{at } R \quad x = 0, -3y \cos \theta = 16 \sin \theta \cos \theta$$

$$y = \frac{16 \sin \theta}{-3}$$

$$\text{ie } R \left(0, \frac{16 \sin \theta}{-3} \right)$$

$$\text{Then coords of } M \left(\frac{8 \cos \theta}{5}, -\frac{8 \sin \theta}{3} \right)$$

$$\text{Subst in } \frac{25x^2}{64} + \frac{9y^2}{64} = 1$$

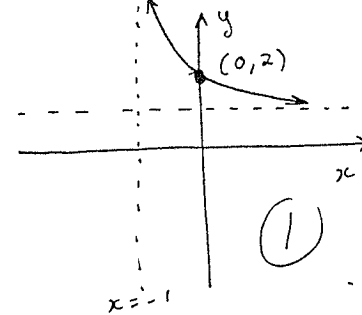
$$\text{LHS} = \frac{25 \times 64 \cos^2 \theta}{64 \times 25} + \frac{9 \times 64 \sin^2 \theta}{64 \times 9} \\ = 1$$

M lies on ellipse

(6)

Question 5

$$1. a) g(x) = 1 + \frac{1}{x+1} \quad x > -1$$

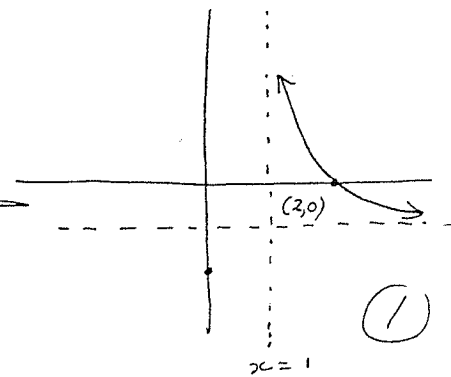


$$b) x = 1 + \frac{1}{y+1} \Rightarrow x-1 = \frac{1}{y+1}$$

(1)

$$g^{-1}(x) = \frac{1}{x-1} - 1$$

$$y+1 = \frac{1}{x-1} \\ y = \frac{1}{x-1} - 1$$



$$c) g(x) = g^{-1}(x)$$

$$1 + \frac{1}{x+1} = \frac{1}{x-1} - 1$$

$$2 + \frac{1}{x+1} = \frac{1}{x-1}$$

$$\frac{2x+3}{x+1} = \frac{1}{x-1}$$

$$(2x+3)(x-1) = x+1$$

$$2x^2 + 3x - 2x - 3 = x + 1$$

$$2x^2 = 4$$

$$x^2 = 2$$

$$x = \pm \sqrt{2}$$

$$\text{But } x > -1 \therefore x = \sqrt{2}$$

(1)

$$2. a) x^3 - 2x^2 + 4x + 2 = 0. \text{ Roots } \alpha, \beta, \gamma.$$

$$\text{Eqn } \alpha - 1, \beta - 1, \gamma - 1 \quad y = x - 1 \quad x = y + 1$$

$$(y+1)^3 - 2(y+1)^2 + 4(y+1) + 2 = 0.$$

$$y^3 + 3y^2 + 3y + 1 - 2y^2 - 4y - 2 + 4y + 4 + 2 = 0.$$

$$y^3 + y^2 + 3y + 5 = 0$$

$$\therefore x^3 + x^2 + 3x + 5 = 0$$

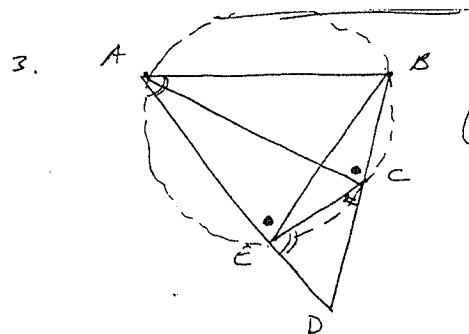
$$b) y = \frac{x}{2} \Rightarrow y = \frac{2x}{2} \Rightarrow x = 2y.$$

$$\therefore (2y)^3 - 2(2y)^2 + 4(2y) + 2 = 0$$

$$8y^3 - 8y^2 + 8y + 2 = 0.$$

$$\therefore \text{eqn } 4x^3 - 4x^2 + 4x + 1 = 0$$

(1)



3. $ABCE$ lie on a circle
 (1) (arc AB subtends equal $\angle$'s at the circumference)
 $\angle AEB = \angle CED$ (data)
 $\therefore \angle BAE = \angle ECD$ (exterior angle of a cyclic quad)
 (1)

4. $x^4 - 2x^3 - 12x^2 + 40x - 32 = 0$ (triple root)
 $4x^3 - 6x^2 - 24x + 40 = 0$ (double root)
 $12x^2 - 12x - 24 = 0$ (single root)

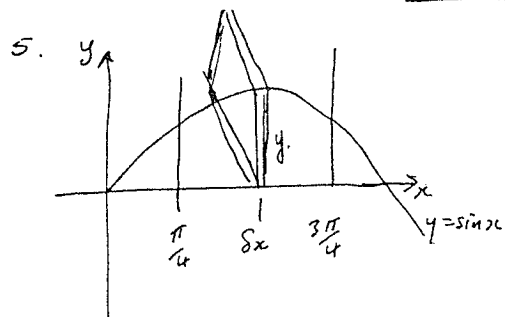
(1) $x^2 - x - 2 = 0$
 $(x+1)(x-2) = 0$

Test $x = -1$ $P(-1) = (-1)^4 - 2(-1)^3 - 12(-1)^2 + 40(-1) - 32 \neq 0$
 $\therefore x = -1$ not a triple root.

(1) Test $x = +2$ $P(2) = 2^4 - 2(2)^3 - 12(2)^2 + 40(2) - 32 = 0$
 $\therefore x = 2$ is triple root.

$\therefore P(x) = (x-2)^3 Q(x) = x^4 - 2x^3 - 12x^2 + 40x - 32$
 $= (x-2)^2(x+a)$ and $-8a = -32 \therefore a = 4$

(1) $\therefore P(x) = (x-2)^3(x+4)$ roots $x = -4, 2, 2, 2$



Area of a slice is y^2
 Volume of a slice
 $SV = y^2 \cdot \delta x$ (1)

Volume of the solid
 $V \approx \sum_{x=\pi/4}^{3\pi/4} y^2 \cdot \delta x$ (1)

$V = \int_{\pi/4}^{3\pi/4} y^2 \cdot dx$

$\therefore V = \frac{1}{2} \int_{\pi/4}^{3\pi/4} (1 - \cos 2x) \cdot dx$

$V = \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_{\pi/4}^{3\pi/4} = \frac{1}{2} \left[\frac{3\pi}{4} - \frac{1}{2} \times -1 \right] - \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \times 1 \right]$

$= \frac{1}{2} \left[\frac{3\pi}{4} + \frac{1}{2} - \frac{\pi}{4} + \frac{1}{2} \right] = \frac{1}{2} \left[\frac{2\pi}{4} + 1 \right] = \frac{\pi}{4} + \frac{1}{2}$

Q6d) $P(x)$ even, monic, degree 4.

(1) $\alpha = 3i \therefore \beta = -3i$ (by rule of conjugates)

Also $\alpha\beta = -9$

(1) But $\alpha\beta = 9 \therefore \beta = -2$
 $\therefore (x^2 + 9)$

But Monic and Even:

$\therefore P(x) = (x^2 + 9)(x^2 - 2)$

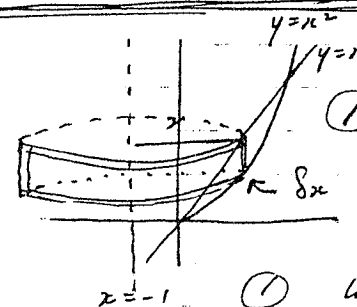
(1) $= (x^2 + 9)(x - \sqrt{2})(x + \sqrt{2})$

(2) $\frac{\cos 15^\circ + \cos 75^\circ}{\sin 15^\circ - \sin 75^\circ} = \frac{2 \cos \frac{15+75}{2} \cos \frac{15-75}{2}}{2 \cos \frac{15+75}{2} \sin \frac{15-75}{2}}$

$= \frac{2 \cos 45^\circ \cos (-30^\circ)}{2 \cos 45^\circ \sin (-30^\circ)}$

(3) $= \frac{1}{\tan(-30^\circ)} = -\frac{1}{\tan 30^\circ} = -\frac{1}{\frac{1}{\sqrt{3}}} = -\sqrt{3}$

(3)



Volume of a Typical Shell

(1) $\Delta V = \pi(R^2 - r^2) \times \text{height}$

$R = 1 + \delta x$

$\therefore \Delta V = \pi((1 + \delta x)^2 - r^2)h$

$\Delta V = \pi(2\delta x + \delta x^2)h$

$x = -1$ (1) $\lim_{\delta x \rightarrow 0} (\delta x)^2$ is negligible

$\therefore \Delta V \approx 2\pi h \cdot \delta x$

Now, $r = 1 + x$ and $h = x - x^2$ (1)

(1) $\therefore V \approx \lim_{\delta x \rightarrow 0} \sum_{x=-1}^0 2\pi h \cdot \delta x =$

(1) $= \int_0^{-1} 2\pi(1+x)(x-x^2) \cdot dx = \int_0^{-1} 2\pi(x-x^3) \cdot dx$

$= 2\pi \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^{-1} = 2\pi \left[\frac{1}{2} - \frac{1}{4} \right] = 0$

$= 2\pi \left[\frac{1}{4} \right] = \frac{\pi}{2}$ c.u.

$$|w|=1 \quad z = \frac{w+7}{1-w}$$

$$z - zw = w + 7$$

$$z - 7 = w + wz \Rightarrow z - 7 = w(1+z)$$

$$\therefore w = \frac{z-7}{z+1} \quad |w|=1$$

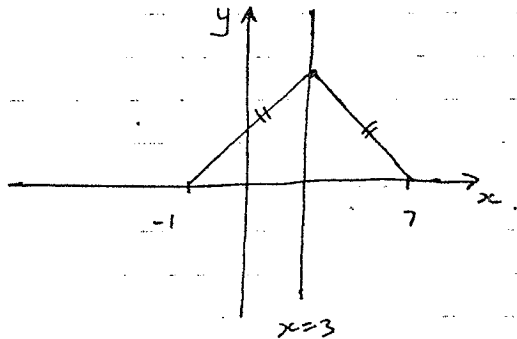
$$\textcircled{1} \quad \therefore 1 = \left| \frac{z-7}{z+1} \right| \Rightarrow |z+1| = |z-7|$$

$$\text{Now } (x+1)^2 + y^2 = (x-7)^2 + y^2$$

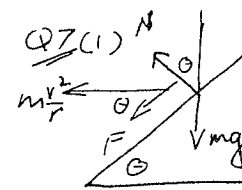
$$x^2 + 2x + 1 + y^2 = x^2 - 14x + 49 + y^2$$

$$16x = 48$$

$$x = 3$$



$\textcircled{1}$



$\textcircled{2}$ (i) Resolving vertically:

$$N \cos \theta = F \sin \theta + mg \quad \text{--- (1)}$$

Resolving horizontally:

$$N \sin \theta + F \cos \theta = m \frac{v^2}{r} \quad \text{--- (2)}$$

$$\textcircled{1} \times \cos \theta \quad \textcircled{2} \times \sin \theta$$

$$N \cos^2 \theta = F \sin \theta \cos \theta + mg \cos \theta$$

$$N \sin^2 \theta + F \sin \theta \cos \theta = m \frac{v^2}{r} \sin \theta$$

$$\therefore N = mg \cos \theta + m \frac{v^2}{r} \sin \theta \quad *$$

$$\textcircled{1} \times \sin \theta \quad \textcircled{2} \times \cos \theta$$

$$N \sin \theta \cos \theta = F \sin^2 \theta + mg \sin \theta$$

$$N \sin^2 \theta \cos \theta + F \cos^2 \theta = m \frac{v^2}{r} \cos \theta$$

$$\therefore F + mg \sin \theta = m \frac{v^2}{r} \cos \theta$$

$$F = m \frac{v^2}{r} \cos \theta - mg \sin \theta \quad *$$

$$\textcircled{3} \textcircled{1} \textcircled{2} \quad 30 \text{ km/hr} \Rightarrow v_1 = \frac{30 \times 1000}{3600} \text{ m/s} \quad 90 \text{ km/hr} = \frac{90 \times 1000}{3600}$$

$$\text{For } v_1 \text{ and } v_2 \quad F_1 + F_2 = 0$$

$$\therefore m \left(\frac{v_1^2}{r} \cos \theta + \frac{v_2^2}{r} \cos \theta \right) = m (g \sin \theta + g \sin \theta)$$

$$\therefore \cos \theta \left(\frac{v_1^2}{r} + \frac{v_2^2}{r} \right) = 2g \sin \theta \quad \begin{matrix} v_1 = \frac{10}{3} \\ v_2 = \frac{5}{3} \end{matrix}$$

$$\therefore \cos \theta = \left(\frac{\left(\frac{100}{9} \right)^2 + \left(\frac{250}{9} \right)^2}{500} \right) = 20 \sin \theta$$

3

$$\therefore \tan \theta = \left(\frac{\left(\frac{25}{3} \right)^2 + \left(\frac{75}{3} \right)^2}{500 \times 20} \right) \Rightarrow \theta = 3^\circ 58'$$

$$\textcircled{ii} \text{ for } F=0 \quad m \frac{v^2}{r} \cos \theta = mg \sin \theta$$

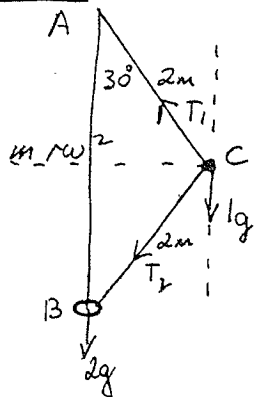
$$v^2 = rg \tan \theta$$

$$v^2 = 500 \times 20 \times \tan \theta$$

$$v = \frac{55.9}{3} \text{ m/s} \div 18.6 \text{ m/s}$$

$$v \div 67 \text{ km/hr}$$

Q Seven (2)



Resolving vertically at C

$$T_1 \cos 30^\circ = T_2 \cos 30^\circ + mg$$

$$\frac{\sqrt{3}}{2} T_1 = \frac{\sqrt{3}}{2} T_2 + mg$$

$$\sqrt{3} T_1 = \sqrt{3} T_2 + 2mg \quad \text{--- (1)}$$

Resolving horizontally at C

$$T_1 \sin 30^\circ + T_2 \sin 30^\circ = m\omega^2 r$$

$$\frac{T_1}{2} + \frac{T_2}{2} = 1 \times 1 \times \omega^2$$

$$\therefore T_1 + T_2 = 2\omega^2 \quad \text{--- (2)}$$

Resolving vertically at B

$$T_2 \cos 30^\circ = 2g \quad \text{--- (3)}$$

$$\therefore T_2 \cos 30^\circ = 3g \quad \text{from (1)}$$

$$\text{Sub in (2)} \quad T_2 = \frac{2g}{\cos 30^\circ} \quad T_1 = \frac{3g}{\cos 30^\circ}$$

$$= \frac{3g}{\cos 30^\circ} + \frac{2g}{\cos 30^\circ} = 2\omega^2 \quad 3$$

$$\frac{5g}{\cos 30^\circ} = 2\omega^2$$

$$\frac{5g}{\frac{\sqrt{3}}{2} \times 2} = \omega^2 \Rightarrow \omega = \sqrt{\frac{5g}{\sqrt{3}}} \text{ rad/sec}$$

$$\omega = 5.373 \text{ rad/sec}$$

(b) Change mass at C to 3 kg and angle to θ .

(i) Vertically at C

$$T_1 \cos \theta = T_2 \cos \theta + 3g$$

Vertically at B

$$T_2 \cos \theta = 2g$$

$$\therefore T_2 \cos \theta = 5g$$

Horizontally at C

$$T_1 \sin \theta + T_2 \sin \theta = m\omega^2 r$$

$$\text{where } r = 2 \sin \theta$$

$$\therefore T_1 \sin \theta + T_2 \sin \theta = 3 \times 2 \sin \theta \times \frac{5g}{\sqrt{3}}$$

$$\therefore \frac{5g}{\cos \theta} + \frac{2g}{\cos \theta} = 6 \times \frac{5g}{\sqrt{3}}$$

$$\frac{7g}{\cos \theta} = \frac{30g}{\sqrt{3}}$$

$$\cos \theta = \frac{7\sqrt{3}}{30}$$

$$\theta = 66^\circ$$

3

(ii) Initially AB

$$= 2 \times 2 \cos 30^\circ = 4 \times \frac{\sqrt{3}}{2}$$

$$3.464 = 2\sqrt{3} \text{ m}$$

Now AB

$$= 2 \times 2 \cos 66^\circ$$

$$= 2 \times \frac{7\sqrt{3}}{30} = 1.617$$

$\therefore$ B will rise

$$3.464 - 1.617$$

$$= 1.85 \text{ m}$$

$$\text{or } 18.5 \text{ cm}$$

Q8(i)

Prove by mathematical induction $x^{2n} - y^{2n}$

is divisible by $(x+y)$ for $n \geq 1$.

Step 1: Prove true for $n=1$

$$x^2 - y^2 = (x-y)(x+y) \quad \text{--- (1)}$$

$\therefore$ true for $n=1$.

Step 2: Assume true for $n=k$ where k is a +ve integer

$$\text{--- (1)} \quad \text{i.e. } x^{2k} - y^{2k} = (x+y)^M \quad \text{where } M \text{ is positive integer}$$

Now: Prove true for $n=k+1$

$$\therefore x^{2(k+1)} - y^{2(k+1)} = x^{2k+2} - y^{2k+2}$$

$$\begin{aligned} \text{--- (1)} \quad &= x^2 \cdot x^{2k} - y^2 \cdot y^{2k} = x^2 \cdot x^{2k} - x^2 \cdot y^{2k} + x^2 \cdot y^{2k} - y^2 \cdot y^{2k} \\ &= x^2(x^{2k} - y^{2k}) - y^2 \cdot y^{2k} + y^2 \cdot y^{2k} - y^2 \cdot y^{2k} \\ &= x^2(x^{2k} - y^{2k}) - y^2 \cdot y^{2k} + y^2 \cdot y^{2k} - y^2 \cdot y^{2k} \end{aligned}$$

$$\begin{aligned} \text{--- (1)} \quad &\text{Now: (from the assumption)} = x^2 \cdot M(x+y) + y^{2k}(x-y)(x+y) \\ &= (x+y)\{x^2 M + (x-y)y^{2k}\} \end{aligned}$$

which is divisible by $(x+y)$.

Step 3 Now the statement is true for $n=k+1$ if true for $n=k$. Statement is true for $n=1$, so is

--- (1) true for $n=1+1=2$ and $n=2+1=3$ and so on for all integer values of n

$$\begin{aligned} \text{(2)} \quad &\frac{1}{x} + \frac{1}{x+a} + \frac{1}{x+b} = 0 \Rightarrow (x+a)(x+b) + x(x+b) + x(x+a) \\ &\Rightarrow 3x^2 + x(2a+2b) + ab = 0 \end{aligned}$$

Roots are $p+q$.

$$\text{--- (1)} \quad \therefore pq = \frac{ab}{3} \quad \text{and } (p+q) = -\frac{2(a+b)}{3}$$

$$\text{Now } (p+q)^2 = p^2 + q^2 + 2pq$$

$$(p+q)^2 = \frac{4(a^2+b^2+2ab)}{9}$$

$$\text{--- (1)} \quad \therefore p^2 + q^2 = (p+q)^2 - 2pq$$

$$= \frac{8ab}{3} - \frac{2ab}{3} = \frac{6ab}{3}$$

$$= 2ab$$

$$\text{--- (1)} \quad \therefore (p+q)^2 = \frac{4(6ab)}{9} = \frac{8}{3}$$

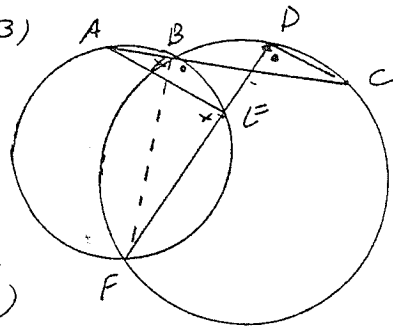
$$\text{and } 2pq = \frac{2ab}{3}$$

$$\text{But } pq = \frac{ab}{3}$$

$$\therefore 6pq = 2ab$$

$$\text{--- (1)} \quad \text{Hence: } p^2 + q^2 = 6pq \quad (\text{both equal } 2ab)$$

Q8(3)



(3)

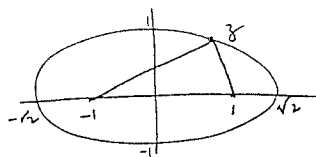
Construction: join BF.

$\angle ABE = \angle AEF$ (L's standing on arc AF)
 $\angle FBC = \angle FDC$ (L's standing on arc FC)

But $\angle ABE + \angle FBC = 180^\circ$ (suppl.)
 $\therefore \angle AEF = \angle FDC$ (suppl. of $\angle AEF$)
 Now $AE \parallel DC$ (pair of alt L's equal)

Q8 4.

$$|z-1| + |z+1| = 2\sqrt{2}$$



$$PS + PS' = 2a$$

$$\therefore a = \sqrt{2}$$

$$(\pm ae, 0) = (\pm 1, 0)$$

$$\therefore ae = 1$$

$$2e = 1$$

$$\therefore e = \frac{1}{2}$$

$$b^2 = a^2(1 - e^2)$$

$$b^2 = 2\left(1 - \frac{1}{4}\right)$$

$$= 2\left(\frac{3}{4}\right)$$

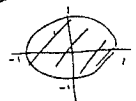
$$b^2 = 1$$

$$\Rightarrow \frac{x^2}{2} + y^2 = 1$$

RTP:

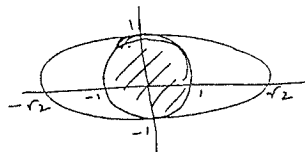
That z represents a pt on or inside the ellipse, $\frac{x^2}{2} + y^2 = 1$

$$\text{now, } |z| \leq 1$$



z represents any pt. on or inside circle

$\therefore$ if z lies on/inside the circle centre (0,0) radius 1, then it also lies on/inside the ellipse $\frac{x^2}{2} + \frac{y^2}{1} = 1$



$$|z-1| + |z+1| \leq 2\sqrt{2}$$

$$|z| \leq 1$$

$$LHS = |(x-1)+iy| + |(x+1)+iy|$$

$$|x^2+y^2| \leq 1$$

$$x^2+y^2 \leq 1$$

$$= \sqrt{(x-1)^2+y^2} + \sqrt{(x+1)^2+y^2}$$

$$= \sqrt{x^2-2x+1+y^2} + \sqrt{x^2+2x+1+y^2}$$

$$= \sqrt{x^2+y^2-2x+1} + \sqrt{x^2+y^2+2x+1}$$

$$\leq \sqrt{1-2x+1} + \sqrt{1+2x+1}$$

$$= \sqrt{2-2x} + \sqrt{2+2x}$$

$$= \sqrt{2} \sqrt{1-x} + \sqrt{2} \sqrt{1+x}$$

$$= \sqrt{2} \left[\sqrt{1-x} + \sqrt{1+x} \right]$$

$$= \sqrt{2} \left[\sqrt{(\sqrt{1-x} + \sqrt{1+x})^2} \right]$$

$$= \sqrt{2} \left[\sqrt{1-x+1+x+2\sqrt{(1-x)(1+x)}} \right]$$

$$= \sqrt{2} \left[\sqrt{2+2\sqrt{1-x^2}} \right]$$

$$= \sqrt{2} \times \sqrt{2} \sqrt{1+\sqrt{1-x^2}}$$

$$= 2 \sqrt{1+\sqrt{1-x^2}}$$

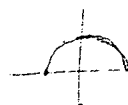
$$\leq 2 \sqrt{1+1}$$

$$= 2\sqrt{2}$$

$$= RHS$$

$$\therefore |z-1| + |z+1| \leq 2\sqrt{2}$$

$$x^2+y^2 \leq 1$$



and $\sqrt{1-x^2} \leq 1$
 when $x=0$
 $\sqrt{1-0^2} = 1$